

Unix ex 1 - Unix Basics

הגשה בזוגות.

נא לציין בשם הקובץ וגם בתוך הדו"ח את מספר תעודות הזהות של שני המגישים.

Log on a Linux machine. Enter your login (user name) and password at relevant prompts.

open terminal.

Enter these commands at the UNIX prompt. For each command copy paste the output and note its role.

- o echo hello world ←
- o date ←
- o hostname ←
- o arch ←
- o uname -a ←
- o uptime ←
- o who am i ←
- o who ←
- o id ←
- o last ←
- o w ←
- o top ← (you may need to press q to quit)
- o echo {con,pre}{sent,fer}{s,ed} ←
- o man ls ← (you may need to press q to quit)
- o man who ← (you may need to press q to quit)
- o clear ←
- o bc -l ← (type quit ← or press Ctrl-d to quit)
- o echo 5+4 | bc -l ←
- o yes please ← (you may need to press Ctrl-c to quit)
- o time sleep 5 ←
- o history ←
- o nano ←

Part 2 Unix File system, Links and quotes

1. Try the following command sequence[copy paste the output into your report]:
 - `cd`
 - `pwd`
 - `ls -al`
 - `cd .`
 - `pwd`
 - `cd ..`
 - `pwd`
 - `ls -al`
 - `cd ..`
 - `pwd`
 - `ls -al`
 - `cd ..`
 - `pwd`
 - `cd /etc`
 - `ls -al | more`
 - `cat passwd`
 - `cd -`
 - `pwd`
2. What is the role of the command `cd -` ? prove your claim by writing a few commands and their output
3. Explore `/dev`. Can you identify what devices are available? Which are character-oriented and which are block-oriented? Can you identify your tty (terminal) device (typing `who` might help); who is the owner of your tty (use `ls -l`)? [write the commands you used to find it out into your report]
4. Make subdirectories called `work` and `play` under your home folder (`~/`).
5. Create a file `lab2.txt` and write “hello world” in your **home directory**. [write the commands you used into your report]
6. Move `lab2.txt` into the subdirectory `play` [write the commands you used into your report]
7. Change into subdirectory `play` and create a symbolic (soft) link called `terminal` that points to your tty device. What happens if you try to make a hard link to the tty device? Why? [write the commands you used into your report for both scenarios]
8. What is the difference between listing the contents of directory `play` with `ls -l` and `ls -lL`? [prove your claim by highlighting a specific output of both commands]

9. Create a file called `hello.txt` that contains the words "hello world".
Copy `hello.txt` to `terminal`. What happens? Why? *[write the commands you used to find it out into your report]*
10. Imagine you were working on a system and someone accidentally deleted the `ls` and `find` commands (by typing: `sudo rm /bin/ls /bin/find`). How could you get a list of the files in the current directory? Try it.
11. How would you create and then delete a file called "`$SHELL`"? Try it. *[write the commands you used to find it out into your report, note the commands used to create this file also]*
12. How would you create and then delete a file that begins with the symbol `#`?
Try it. *[write the commands you used to find it out into your report]*
13. How would you create and then delete a file that begins with the symbol `-`?
Try it. *[write the commands you used to find it out into your report]*
14. Still in your home directory, copy the entire directory `play` to a directory called `work`, preserving the symbolic link.
15. Delete the `work` directory and its contents with one command. Accept no complaints or queries. *[write the commands you used to find it out into your report]*
16. Experiment with the options on the `ls` command. What do the `-d`, `-R`, `-F`, `-i` options do?